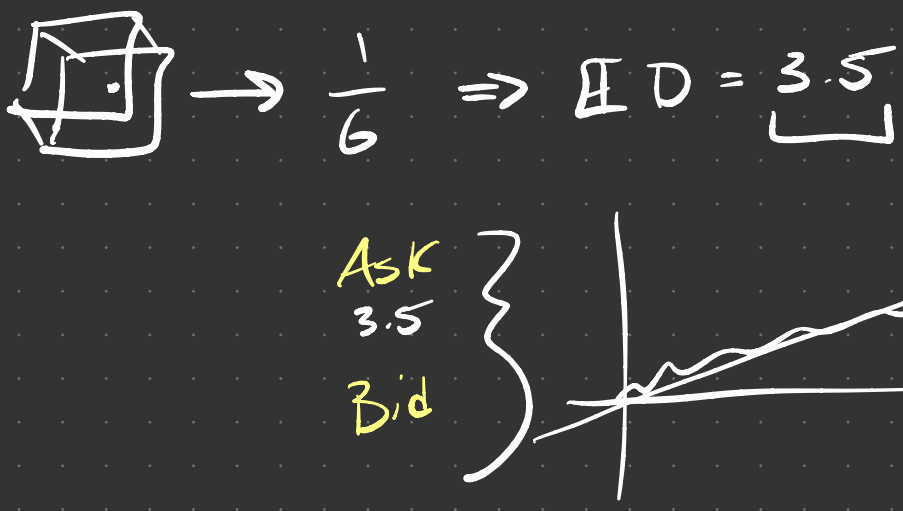


Defn: Random Experiment

A procedure or process whos outcome can't be predicted

- Coin
- Dice roll
- Roulette

Defn: Probability Model

A probability model is a math description of a Random experiment that specifies outcomes and assigns probabilities to different "events"

- What outcomes are possible?
- How likely is each event?

- How does knowing one event impact the prob. of another
- What is the exp. value of a random qty?

Defn: Sample Space

The set of all possible outcomes of a Random experiment

e.x.) Sample Space Dice Roll

$$\Omega = \{1, 2, 3, 4, 5, 6\}$$

$$\Omega^2 = \Omega \otimes \Omega = \{(1,1), (1,2), \ldots$$
$$(2,1), (2,2), \ldots$$
$$\vdots$$
$$(6,1), (6,2), \ldots\}$$

Defn: Event

An event is a subset of the sample space.

$$\Omega = \{1, 2, 3, 4, 5, 6\}$$

"Even Dice Rolls" $= E = \{2, 4, 6\}$

$$E \subseteq \Omega$$

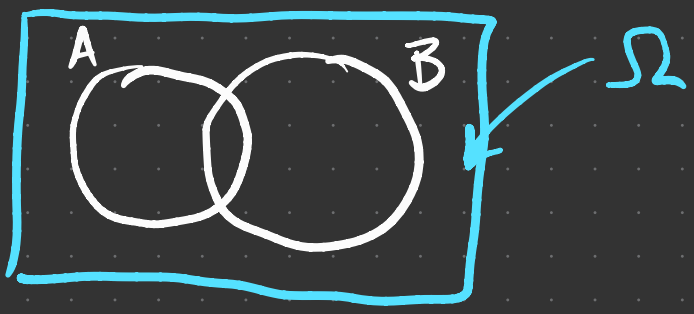
Defn: Operations on Sets

Given two events $A, B \subseteq \Omega$

Union: $A \cup B = U$

Intersection: $A \cap B = I$

Complement: $A^c = \Omega \setminus A$



Defn: Probability Measure

A probability measure, P, is a function that assign a number to each event and it satisfies the following axioms:

Axiom 1: Nonnegativity

$$P(A) \geq 0$$

Axiom 2: Normalization

$$P(\Omega) = 1$$

Axiom 3: Countable Additivity



pairwise mutually exclusive events

$\forall i, j \quad A_i \cap A_j = \emptyset$

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i)$$

$$0 \leq P(A) \leq 1 \quad , \quad \forall A$$

Rule 1: Complement Rule

$$P(A^c) = 1 - P(A)$$

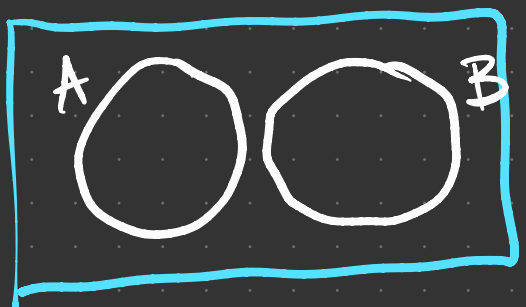
$$P(A^c) + P(A) = 1$$

$$A^c \cup A = \Omega$$

Rule 2: Addition Rule

For any two events A, B

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$



$P(A \cap B) = 0$

$A \cap B = \emptyset$

Defn) Equally Likely Outcomes

A sample space has equally likely outcomes if every event has the same probability.

Ex) N outcomes, each has prob. $\frac{1}{N}$

$$P(A) = \frac{\text{Number of favorable outcomes}}{\text{The total \# of outcomes}}$$

E - "Even dice roll" $= \{2, 4, 6\}$

$$P(E) = \frac{|E|}{|\Omega|} = \frac{3}{6} = \frac{1}{2}$$

Ex) Roll two dice independently.

S - "At least one six"; $P(S) = \frac{|S|}{|\Omega|} = \frac{11}{36}$

$|\Omega| = 36$, $|S| = 11$

$$\frac{25}{36} = P(S^c) \Rightarrow P(S) = 1 - P(S^c)$$

$$= 1 - \frac{25}{36} = \frac{11}{36}$$

Combinations + Permutations

Permutation Formula

$$P(n,r) = \frac{n!}{(n-r)!} \qquad \begin{matrix} n=6 \\ r=3 \end{matrix}$$

$$\frac{6!}{(6-3)!} = \frac{6!}{3!} = \frac{6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{3 \cdot 2 \cdot 1}$$

Combinations Formula

$${}_nC_r = \binom{n}{r} = \frac{n!}{r!(n-r)!}$$

A B C D E F

6

$${}_6C_3 = \binom{6}{3} = \frac{6!}{3!(3)!} = \frac{120}{3!}$$

$$= \frac{120}{6} = 20$$

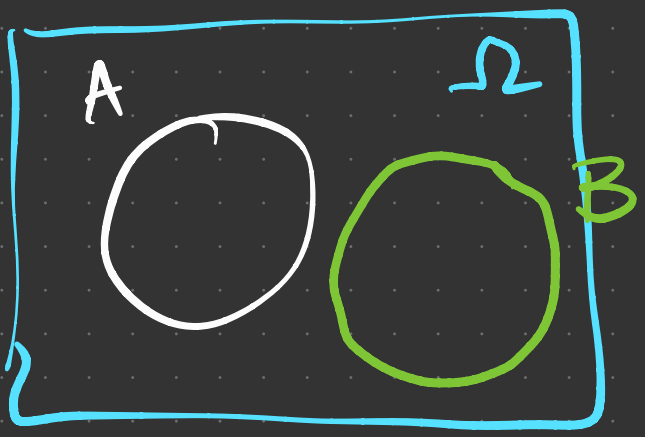
Defn: Conditional Probability

Given events A, B

The cond. prob A given B occured

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$P(B) > 0$



Multiplication Rule

$$P(A \cap B) = P(A|B)P(B)$$

Defn: Partition

A collection of events $B_1, B_2, \ldots$ forms a partition if

1) $\forall i,j \; B_i \cap B_j = \emptyset$

2) $\bigcup_{i=1}^{\infty} B_i = \Omega$

Law of Total Probability

If $B_1, B_2, \ldots$ form a partition

$$P(A) = \sum_{i=1}^{\infty} P(A|B_i)P(B_i)$$

Bayes' Theorem

For a partition $B_1, B_2, \ldots$

$$P(B_i|A) = \frac{P(A|B_i)P(B_i)}{\sum_{j=1}^{\infty} P(A|B_j)P(B_j)}$$

$$P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{P(A|B)P(B)}{P(A)}$$